

A PROJECT REPORT ON

**Optimizing Crop Selection And Profitability Through
Machine Learning-Based Recommendations**

**SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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OF

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SUBMITTED BY

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CERTIFICATE

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ABSTRACT

In modern agriculture, the use of data-driven techniques has become essential for optimizing crop selection to maximize yield and profitability. This project presents the development of a web-based crop recommendation service that integrates environmental factors such as soil composition, temperature, humidity, and rainfall to suggest the most suitable crops for a given region. Unlike traditional crop recommendation systems, this solution also considers the profitability of crops by providing alternative suggestions that align with current market trends. Using machine learning algorithms and real-time environmental data, the system enhances decision-making for farmers, helping them to optimize both crop productivity and financial outcomes. The platform is designed to be user-friendly, offering accessible and actionable insights that aim to boost agricultural efficiency and sustainability. Future enhancements may include incorporating additional economic data, real-time market analysis, and expanded datasets to further refine crop recommendations.

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List of Abbreviations

API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CSV	Comma-Separated Values
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
FAQ	Frequently Asked Questions
GCP	Google Cloud Platform
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
HTML	HyperText Markup Language
JWT	JSON Web Token
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
RBF	Radial Basis Function
RoBERTa	Robustly Optimized BERT Approach
SDLC	Software Development Life Cycle
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
UI	User Interface
UMAP	Uniform Manifold Approximation and Projection
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 Overview

In the modern agricultural landscape, optimizing crop selection to maximize yield and profitability is a growing challenge for farmers. With the increasing availability of environmental data, there is significant potential to enhance decision-making through data-driven approaches. Our project aims to address this challenge by developing a web-based crop recommendation system that integrates critical environmental factors such as soil composition, temperature, humidity, and rainfall to suggest the most suitable crops for a given region. The system leverages machine learning algorithms to analyze real-time data and provide tailored recommendations that consider both crop productivity and market trends. Unlike traditional systems, this approach also factors in the economic aspects of farming, offering alternative crop options that align with current market demands. The goal is to help farmers make informed decisions that boost agricultural efficiency, profitability, and sustainability. In the future, we plan to incorporate additional economic data, real-time market analysis, and expanded datasets to refine and personalize crop recommendations even further, ultimately creating a more dynamic and responsive agricultural ecosystem.

1.2 Motivation

In modern agriculture, farmers face the challenge of selecting the most suitable crops for varying environmental conditions while also considering profitability. Traditional crop recommendation systems often focus solely on environmental factors like soil type and weather, without incorporating economic variables such as market demand and crop prices. This limitation can lead to suboptimal decisions, affecting both crop yield and financial outcomes. Farmers need a more efficient and data-driven approach to navigate these complexities and maximize their agricultural potential. To address this challenge, we aim to develop a web-based crop recommendation system that not only analyzes environmental factors but also incorporates market trends and profitability data. By combining real-time data analysis with machine learning, this system will help farmers make informed decisions that optimize both productivity and financial returns. This solution will ultimately empower farmers to adapt to changing conditions, improve sustainability, and enhance overall agricultural efficiency.

1.3 Problem Definition and Objectives

In modern agriculture, farmers often face the challenge of selecting the right crops to grow based on a combination of environmental factors and market trends. Traditional crop recommendation systems typically focus on factors such as soil composition, weather conditions, and irrigation, but they often fail to incorporate the economic aspects, such as market demand and profitability. This gap results in suboptimal crop decisions, which can impact both agricultural productivity and financial returns. Additionally, existing systems do not offer personalized or dynamic recommendations, leaving farmers to navigate an overwhelming amount of data without clear guidance. Currently, farmers rely on manual decision-making or generalized recommendations that fail to account for real-time conditions or market fluctuations. This approach can lead to inefficiencies in crop selection, as farmers may choose crops that are not well-suited to current environmental conditions or market demand, resulting in reduced yields or financial losses. The primary objective of this project is to address these challenges by developing a web-based crop recommendation system that integrates environmental data (such as soil composition, temperature, humidity, and rainfall) with market data to provide personalized, data-driven recommendations. The system will use machine learning algorithms to analyze real-time data and offer recommendations that optimize both crop yield and profitability. By doing so, the system aims to empower farmers to make informed decisions that improve agricultural efficiency, increase financial returns, and promote sustainability in farming practices.

1.4 Project Scope & Limitations

The proposed project aims to enhance the decision-making process for farmers by providing a web-based crop recommendation system that analyzes environmental and market data. The system leverages machine learning algorithms to suggest the most suitable crops for a given region based on factors such as soil composition, temperature, humidity, and rainfall. Additionally, the system incorporates profitability analysis to suggest alternative crops aligned with current market trends, helping farmers make financially viable decisions. Key features of the system include accurate crop recommendations powered by machine learning, real-time data integration, and a user-friendly dashboard that allows farmers to view and manage recommendations efficiently. The scope of the project focuses on providing data-driven recommendations for a broad range of crops and environmental conditions, specifically targeting farmers looking to optimize yield and profitability. However, the project does face limitations in terms of data availability and variability. The accuracy of the recommendations depends heavily on the quality and comprehensiveness of the environmental and market data used. The system may also be less adaptable to highly specialized agricultural contexts or unique regional conditions that require more tailored solutions. Furthermore, the project is designed for general use and may require additional customization to meet the needs of specific crops or farming practices, potentially limiting its applicability in niche farming scenarios.

1.5 Methodology of Problem solving

The proposed system is a web-based crop recommendation platform designed to assist farmers in making data-driven decisions regarding crop selection. The system leverages machine learning algorithms to analyze various environmental and market data to recommend the most suitable crops based on factors like soil composition, temperature, humidity, and rainfall. It also incorporates profitability analysis, taking current market trends into account to suggest financially viable crops. The solution aims to optimize both crop yield and farmer profitability while enhancing decision-making efficiency.

The system will be accessible through a user-friendly web interface, allowing farmers to input relevant data about their farm's location, soil characteristics, and environmental conditions. Once registered, the platform will analyze the input data using machine learning algorithms, such as Random Forest and Support Vector Machines (SVM), to generate crop recommendations that are tailored to the specific conditions of the farm. Additionally, the system will incorporate real-time market data to recommend crops that are currently profitable, offering farmers alternative suggestions that align with market demand.

The recommendation engine will continuously refine its predictions by learning from historical data and user feedback, improving its accuracy over time. To facilitate ease of use, the system will feature an intuitive dashboard where farmers can view recommendations, track market trends, and monitor environmental factors. Furthermore, the platform will provide notifications and alerts to keep farmers informed about potential issues such as changing weather patterns or market fluctuations. This will enable farmers to make proactive decisions and optimize their agricultural practices, ultimately leading to increased crop yield and financial success.

CHAPTER 2

LITERATURE SURVEY

Effective crop recommendation systems play a crucial role in assisting farmers with informed decision-making, ultimately optimizing crop yield and sustainability. With the growing complexity of environmental data, such as soil composition, temperature, humidity, and rainfall, it becomes challenging to manually assess the most suitable crops for different regions. Machine learning (ML) and data-driven techniques have revolutionized this process by automating the analysis of large datasets, offering personalized crop suggestions based on a variety of factors. ML algorithms, such as Random Forest, Support Vector Machines (SVM), and Gradient Boosting, are employed to predict the best crop options, while considering both environmental factors and market dynamics. These advancements in ML and natural language processing (NLP) have led to more accurate, real-time crop recommendations, helping farmers improve productivity, reduce risks, and adapt to changing environmental conditions. The following literature survey reviews various approaches and methodologies used in crop recommendation systems, highlighting the use of machine learning algorithms in enhancing agricultural decision-making.

In the development of crop recommendation systems, one study by Shivbhakta et al. (2024) proposes a crop prediction system that utilizes weather and soil data for decision-making. Various machine learning algorithms, including Gradient Boosting, were employed, with Gradient Boosting achieving the highest accuracy in crop prediction. The research emphasizes the importance of environmental factors, such as temperature, humidity, and rainfall, in determining suitable crops. This approach highlights how combining both weather and soil data can significantly improve crop selection, leading to enhanced productivity and optimized agricultural outputs for farmers [1].

Another study by Bandara et al. (2020) explores a crop recommendation system using real-time environmental data collected through sensors, with Naïve Bayes and SVM algorithms applied for prediction. This system achieved over 92% data integration and offers advantages for systems aiming to dynamically adjust recommendations based on up-to-date environmental factors. The research demonstrates how sensor-based crop recommendations can aid farmers in making informed decisions and provide sustainable, high-yielding solutions tailored to their specific environmental conditions [2].

In an attempt to assist Indian farmers, Patel et al. (2024) investigate the use of machine learning for selecting crops based on climatic, soil, and geographical factors. Decision Trees and Naïve Bayes were among the most successful algorithms, achieving over 99% accuracy and reduce crop failure risks. The findings underline the significance of localized data in crop selection, ensuring that farmers make optimal decisions based on environmental compatibility, thus improving productivity and minimizing losses due to adverse conditions [3].

A different study by Suguna et al. (2024) investigates the use of MultiLayer Perceptron (MLP) and Linear Regression (LR) for recommending crops based on soil, climatic, and geographical data. It incorporates seasonal data segmentation for time-specific recommendations, achieving an accuracy of 81% and the seasonal nature of agriculture to optimize yield potential. By incorporating the timing of planting into the recommendation system, farmers can receive better-targeted advice that aligns with the optimal growing periods for specific crops, improving productivity and aligning with seasonal variations [4].

Focusing on Random Forest, a study by Kale and Mohapatra (2024) explores its use in crop recommendation systems and finds it achieves an accuracy rate of 99.22% using weather patterns, to suggest the best crops. Random Forest's ability to handle large datasets and make accurate predictions makes it an ideal choice for crop recommendation systems. This study provides valuable insight into using Random Forest as a robust algorithm that can guide decision-making in agricultural practices, ensuring better crop selection based on environmental compatibility [5].

A further study by Pande et al. (2021) investigates the optimization of crop recommendation systems by combining Support Vector Machine (SVM) with Particle Swarm Optimization (PSO). This hybrid approach achieved 98.64% machine learning performance. By fine-tuning the SVM model using PSO, the system provided more accurate crop recommendations, making it highly effective for dynamic agricultural environments. The research highlights the potential of combining machine learning models with optimization algorithms to further improve the reliability and precision of crop recommendation systems in diverse farming conditions [6].

The AgriRec system, proposed by Patel and Patel (2023), presents a multi-criteria approach by recommending both crops and fertilizers based on soil properties, water levels, and market value. The research highlights the integration of economic factors alongside environmental conditions, ensuring that crop recommendations also consider financial viability. This approach offers farmers not only the most suitable crops but also the best fertilizers to improve yield. By incorporating market data, the system provides a holistic recommendation process that aligns with both environmental sustainability and economic profitability, making it highly relevant for maximizing farm productivity and profitability [7].

In another study, Shireesha et al. (2024) developed a web-based application for crop recommendation using machine learning algorithms such as Random Forest and XGBoost, complemented by deep learning models like ResNet-9 for disease detection. The platform not only recommends crops but also suggests fertilizers and detects diseases from plant images, creating a comprehensive decision-support tool. This integrated approach ensures that farmers receive accurate recommendations, not just for crop selection, but for overall crop management, including disease prevention and nutrient optimization, which could improve both yield and sustainability [8].

A review by Dahiphale et al. (2023) examines various crop recommendation systems and the use of different machine learning algorithms, including Random Forest, which achieved an accuracy of 99.5%. The study emphasizes how data-driven recommendations improve agricultural productivity. By focusing on localized data such as weather patterns, soil composition, and seasonal trends, the study highlights the role of machine learning in providing accurate, scalable, and real-time decision support for farmers to optimize crop selection and maximize yield [9].

Table 2.1: Literature Survey

Title	Work Done
Yield Prediction	Implemented machine learning models such as Linear Regression and Random Forest to predict crop yield based on features including soil type, rainfall, temperature, humidity, and previous yield data.
Crop Recommendation	Developed a recommendation system that integrates soil data (NPK levels), weather parameters, and market trends. Utilized classification algorithms like Decision Trees, K-Nearest Neighbors, and SVM to suggest optimal crops for given conditions.
Weather Forecast	Integrated OpenWeatherMap API to fetch real-time and forecasted weather data, including temperature, humidity, rainfall, and wind speed.
Weather Report	Designed a weather reporting module that offers real-time climate updates with graphical visualizations (charts for temperature, rainfall trends, and humidity levels).
Registration and Login	Created secure user authentication system using Flask and Sql. Enabled login and signup functionalities with email/password. Applied encryption for password protection and ensured session management for personalized user dashboards.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- Farmers using the platform will provide accurate and up-to-date information about their farm's environmental conditions, such as soil type, temperature, humidity, and rainfall.
- Users will engage with the platform ethically, providing relevant and truthful input to ensure accurate crop recommendations.
- Farmers will regularly check the platform for crop suggestions, market trends, and updates related to weather conditions or other relevant alerts.
- The system will assume that users have basic knowledge of agriculture and technology to navigate the web-based interface effectively.
- The machine learning algorithms will function optimally, providing crop recommendations based on the available environmental and market data.
- Market data, such as crop prices and demand trends, will be updated regularly to ensure the accuracy of profitability predictions.
- The accuracy of crop recommendations will depend on the quality and completeness of the input data provided by farmers, including soil and weather conditions

3.1.2 Dependencies

- Farmers must have access to a device (smartphone, tablet, or computer) with an internet connection to interact with the platform and input necessary data.
- The platform's performance will rely on a stable and high-speed internet connection to process real-time data and provide accurate recommendations.
- The successful operation of the machine learning models depends on the availability of large, high-quality datasets for training and validation.
- Integration of third-party APIs for weather forecasting, market price tracking, and other external data sources will be necessary to provide real-time and accurate information.
- The system's performance will depend on adequate server resources to process large amounts of environmental and market data, as well as to generate recommendations efficiently.
- Farmers using the platform must have access to basic technical training or support to ensure proper use of the system and its features.

3.2 Functional Requirements

The following are the functional requirements of the crop recommendation system:

3.2.1 User Registration and Profile Creation

A secure registration process should be implemented, allowing farmers to create their profiles on the platform. Users must provide essential information, such as farm location, soil type, and crop preferences, which will be verified by the system to ensure accuracy and relevance.

3.2.2 Crop Recommendation Interface

The platform should feature a user-friendly interface displaying recommended crops based on environmental factors, soil data, and market trends. The interface should allow farmers to view details about each recommended crop, including profitability estimates, growth requirements, and potential market demand.

3.2.3 Automated Notifications

The system should provide timely notifications related to weather changes, new crop recommendations, and market updates. These notifications will help farmers stay informed and make real-time adjustments to their crop planning based on current environmental and economic conditions.

3.2.4 Environmental Data Analysis and Clustering

The platform should use machine learning algorithms to analyze and cluster environmental data, such as temperature, rainfall, and soil composition, to recommend suitable crops. This functionality will enable farmers to receive crop suggestions tailored to their specific farm conditions.

3.2.5 Profitability Analysis

To assist farmers in maximizing financial outcomes, the system should analyze current market trends and project profitability for each recommended crop. This analysis should consider factors like local demand, crop prices, and production costs, providing farmers with insights into the most profitable options.

3.2.6 Market Trend Analysis

The platform should incorporate market trend analysis to identify crops with high potential demand based on local and regional market data. This feature will help farmers choose crops that align with current market needs, enhancing the likelihood of successful sales and profitability.

3.2.7 Analytical Dashboard

A comprehensive dashboard should be available for farmers to track key metrics such as crop profitability, weather forecasts, soil conditions, and market trends. This dashboard should offer visual insights to help farmers make data-driven decisions, optimizing their crop selection and overall agricultural productivity.

3.3 External Interface Requirements

The proposed crop recommendation system's interfaces are categorized into three main sections: web-based interface, mobile application, and API services.

3.3.1 Web Based Interface

The web-based dashboard will serve as the primary interface for farmers and administrators to interact with the platform. This interface will be accessible through any major web browser, such as Google Chrome, Firefox, or Safari. Users will be able to register, update their farm profiles, view recommended crops, and access environmental and profitability insights. The dashboard will be developed using modern JavaScript frameworks like React or Angular, ensuring a responsive, user-friendly experience that adapts to different screen sizes.

3.3.2 Mobile Application

To provide farmers with a more convenient way to access the platform, a mobile application will be developed for Android and iOS devices. This app will offer core functionalities, such as viewing crop recommendations, receiving weather and market alerts, and accessing profitability analysis. The app will be designed for offline capabilities to support farmers in areas with limited internet connectivity, allowing for data synchronization when a connection is available.

3.3.3 API Services

The system will provide RESTful API services for seamless integration with third-party agricultural and market data sources. These APIs will allow developers and agricultural organizations to programmatically access core features, such as environmental data processing, crop recommendation, and profitability analysis. Detailed API documentation will be available to facilitate integration and support custom applications or services that require real-time crop and market information.

3.4 Non-functional Requirements

3.4.1 Performance Requirements

1. User Satisfaction: The system should deliver a reliable and user-friendly experience for farmers and administrators, meeting their expectations for ease of use, accessibility, and efficiency.

2. 2. Average Response Time: The system should process user requests and deliver crop recommendations within an average response time of under 5 seconds, allowing users to quickly access critical information.
3. 3. Reliability: The platform must maintain consistent reliability by providing accurate crop recommendations and environmental data, minimizing downtime and ensuring uninterrupted service.
4. 4. System Availability: The platform should be available to users at least 99system when they need it most.

3.4.2 Safety and Security Requirements

1. 1. User Account Security: User accounts and associated data must be securely stored using encryption and strict access protocols to prevent unauthorized access and protect sensitive information.
2. 2. Password Integrity: The system will enforce strong password policies, with an option for two-factor authentication to enhance account security and prevent unauthorized access.
3. 3. Data Protection: The platform will comply with data protection regulations such as GDPR, ensuring user data is securely handled and that users are informed about data usage and privacy policies.
4. 4. API Security: The system's APIs must implement secure authentication (e.g., OAuth 2.0) to prevent unauthorized access, ensuring that only authenticated users and approved applications can access data and system functionalities.

3.5 System Requirements

3.5.1 Database Requirements

- MySQL

3.5.2 Software Requirements

- Backend APIs: Nodejs, Expressjs , Flask
- Frontend: ReactJS, Bootstrap
- Containerization: Docker
- Deployment : Microsoft Azure
- Operating System : Windows/Linux

3.5.3 Libraries and Tools:

- **Scikit-learn:** For implementing machine learning algorithms for crop recommendations (e.g., Decision Tree, Random Forest) and clustering techniques.
- **TensorFlow / PyTorch:** For training and deploying advanced machine learning models when needed.
- **Pandas and NumPy:** For data preprocessing and efficient handling of environmental data and user inputs.
- **Flask:** To manage backend services and integrate ML models for the crop recommendation process.
- **MySQL Database:** For structured data storage related to user interactions, crop suggestions, and environmental factors.

3.5.4 Hardware Requirements

- Processor: intel i5 or higher
- Memory: Minimum 8GB ; 16GB recommended
- Storage: SSD with at least 256 GB for model storage and data processing
- GPU: Dedicated GPU for faster training of deep learning models

3.6 Analysis Models: SDLC Model

The Agile methodology will be the foundation for our project development approach, emphasizing flexibility, adaptability, and user-centric progress. Agile's iterative nature allows for continuous enhancement and adaptation throughout the project's life cycle, making it ideal for developing dynamic solutions like the social media interaction enhancement system.

3.6.1 Agile Model for SDLC

In this project, we adopted the Agile methodology and implemented its principles to build the system step-by-step. The development process was organized into iterative cycles known as sprints, with each sprint addressing a specific feature set. This allowed us to promptly act on feedback and continuously refine the system. Each sprint encompassed the following stages:

1. **Plan:** In this initial phase, we collected detailed requirements related to comment clustering, query matching, and sentiment analysis integration. The emphasis was on understanding user expectations and prioritizing features that provided maximum value to content creators.

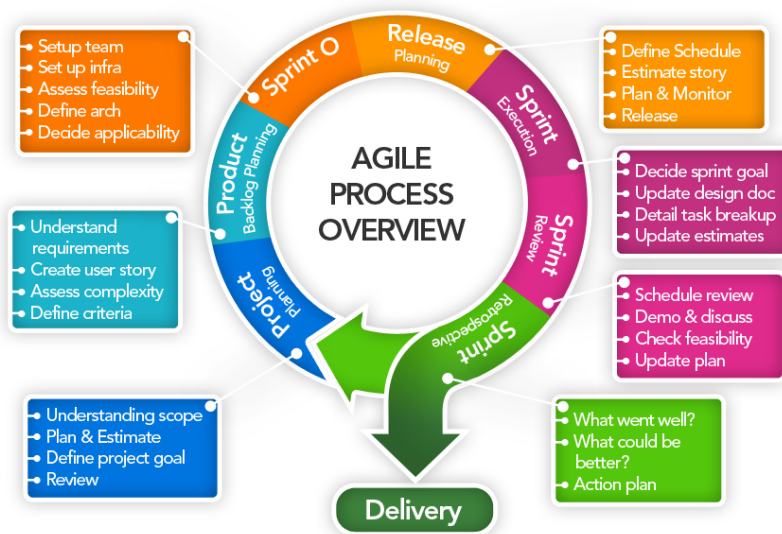


Figure 3.1: Agile Model for SDLC

2. **Design:** During this stage, we developed architectural diagrams, UI prototypes, and data flow representations for clustering techniques, query matching operations, and sentiment classification. We also planned the integration of third-party APIs and outlined the system's backend structure.
3. **Develop:** This phase involved writing the code and implementing the core modules, such as intelligent comment clustering, query handling, and sentiment analysis. We made use of modern tools and best practices to ensure the system was both scalable and efficient.
4. **Test:** In this phase, we rigorously tested the modules using a mix of manual and automated approaches. The objective was to verify the accuracy of clustering, evaluate the responsiveness of query matching, and ensure the sentiment analysis correctly interpreted user comments.
5. **Release:** Once all components passed the testing phase, we deployed them to a staging environment for final evaluations. This step also included setting up server configurations, managing the database, and confirming seamless integration with social media platforms.
6. **Feedback:** At the end of every sprint, we collected feedback from stakeholders such as content creators and technical reviewers. This feedback guided our feature improvements, bug fixes, and sprint planning for subsequent development cycles.

By following the Agile methodology, the project was able to evolve based on ongoing feedback, resulting in a more user-responsive and need-driven solution for content creators. The incremental development approach helped us consistently enhance the system, ultimately delivering a robust tool for improving interactions on social media platforms.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

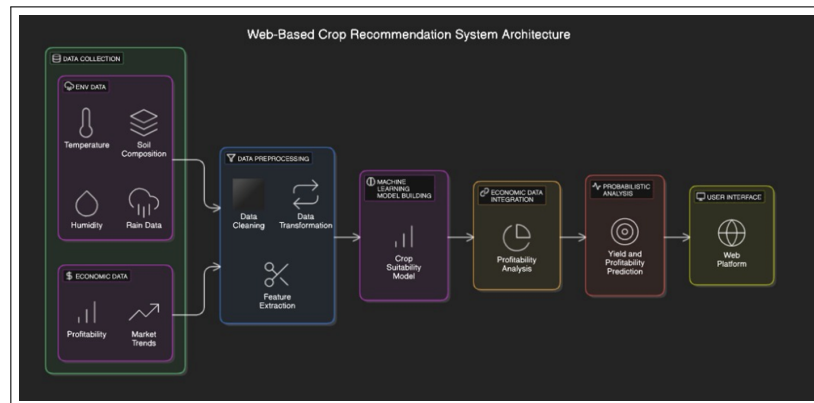


Figure 4.1: System Architecture

The System Architecture for the crop recommendation platform consists of several core components that work together to analyze environmental factors, crop profitability, and suggest the most suitable crops for a specific region. Below is a detailed description of each component in the architecture:

- **Data Collection Module:**

The system begins by gathering environmental and agricultural data from various sources, such as real-time sensors, weather APIs, and agricultural databases. This data includes soil composition, weather parameters (temperature, humidity, rainfall), and regional crop data. The collected data is temporarily stored before entering the preprocessing stage to ensure its suitability for model analysis.

- **Data Preprocessing Pipeline:**

Once the data is collected, it flows through a preprocessing pipeline. This module cleans and standardizes the data by handling missing values, encoding categorical variables, and normalizing numerical values. This step is essential to enhance data quality, making it ready for accurate model predictions and probabilistic analysis.

- **Feature Store:**

The feature store acts as a structured repository for processed data, storing essential features such as soil nutrient levels, climatic conditions, and historical crop yields. This repository supports the machine learning models by supplying a consistent set of features for training, validation, and real-time predictions.

- **Machine Learning Model Building Module:**

This module constructs predictive models to recommend the most suitable crops based on environmental factors and historical data. The models utilize machine learning algorithms like Random Forest, Decision Tree, and Gradient Boosting to identify optimal crop options. Each model is trained and validated using the data stored in the feature store to ensure high accuracy in crop recommendation.

- **Economic Data Integration Module:**

To provide additional value to the recommendation system, this module integrates real-time economic data such as market prices, demand trends, and profitability forecasts. By incorporating these economic factors, the system can suggest crops that are not only suited to environmental conditions but also have high market potential, thus enhancing profitability for farmers

- **Probabilistic Analysis Module:**

This module uses probabilistic algorithms to assess the likelihood of crop success based on various scenarios, such as seasonal changes or shifting market conditions. This analysis helps provide alternative recommendations under uncertain or varying conditions, giving users greater flexibility in decision-making.

- **User Interface (UI):**

The user interface is a web-based dashboard designed for ease of use, allowing farmers and agricultural advisors to access crop recommendations, economic insights, and data trends through a clear and intuitive layout. Built using React.js, this interface enables users to configure inputs, view recommendations, and adjust system parameters to suit specific farming needs.

This architecture empowers farmers with actionable insights by integrating real-time environmental and economic data, streamlining the decision making process, and recommending the most profitable crops suitable for their land and conditions.

4.2 Data Flow Diagram

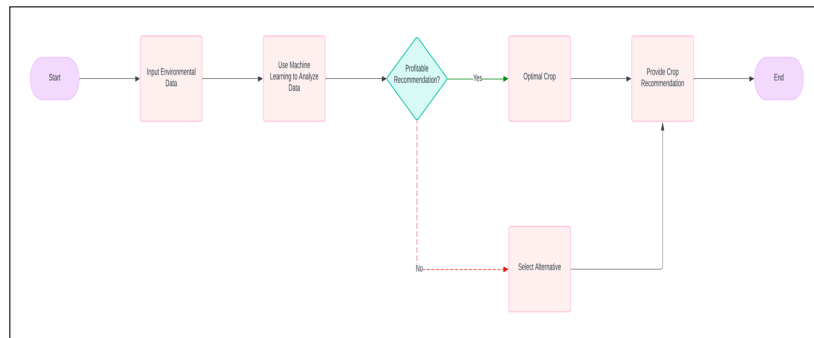


Figure 4.2: Data Flow Diagram

4.3 Entity Relationship Diagram

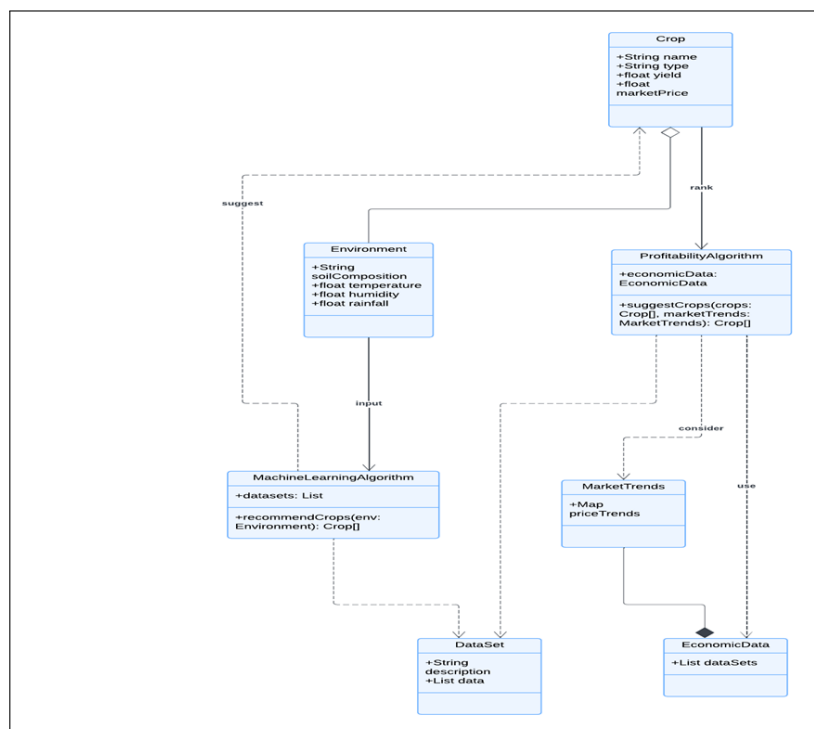


Figure 4.3: ER Diagram

4.4 Use Case Diagram

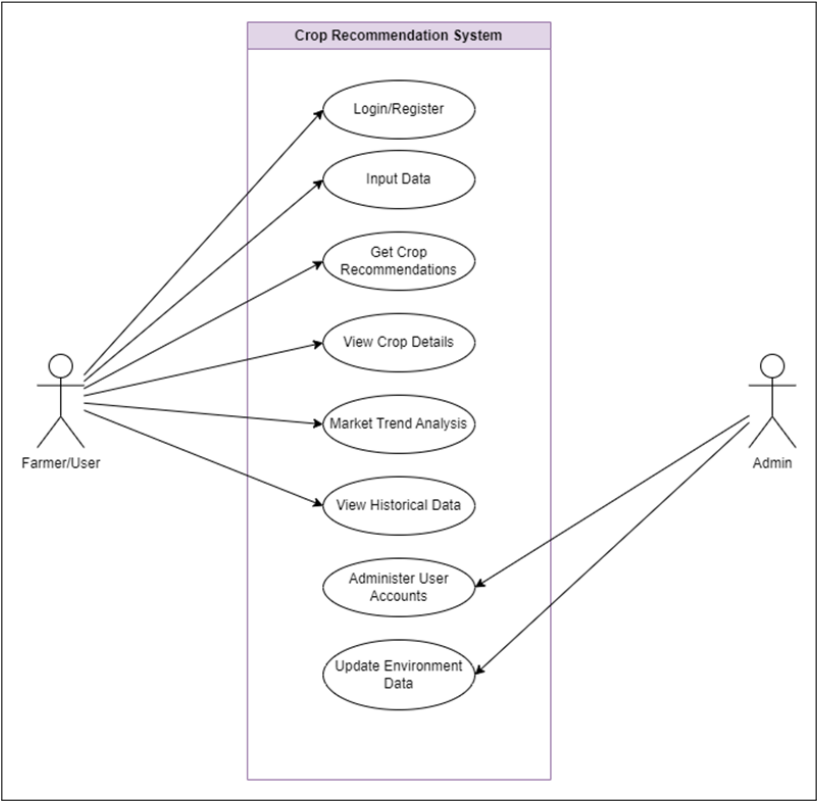


Figure 4.4: Use Case Diagram

4.5 Sequence Diagram

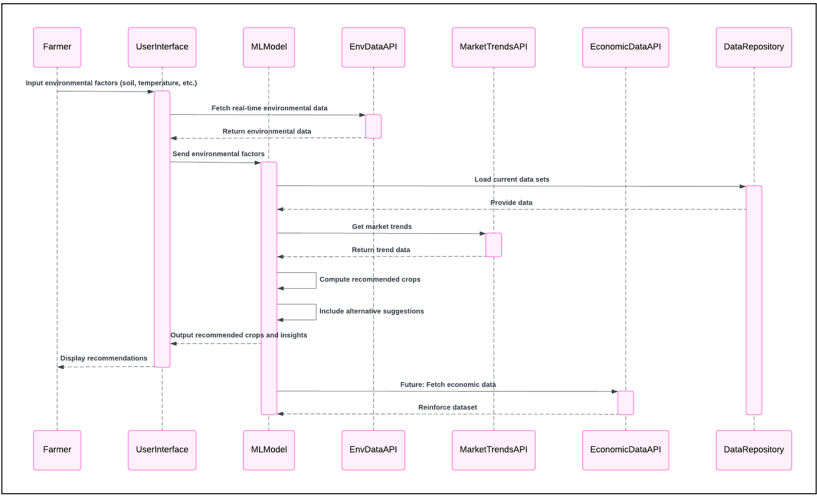


Figure 4.5: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Research and Development:** Estimated a total of 5 months for researching and developing the core features of the crop recommendation system. This phase includes gathering and analyzing environmental data, selecting and implementing machine learning models for crop predictions, and integrating external data sources like weather APIs and IoT devices. The team will explore various algorithms for predicting crop suitability and profitability based on diverse environmental factors.
- **Testing and Evaluation:** Estimated 2.5 months for comprehensive system testing. This will include unit testing for individual components (such as data processing, machine learning models, and IoT sensor integration) and integration testing to ensure that all parts of the system work seamlessly together. The testing phase will also include performance optimization, security assessments, and user feedback testing to evaluate the system's real-world applicability and accuracy.
- **Documentation and Review:** Estimated 1 month for developing thorough project documentation. This will include system architecture documentation, user guides for farmers, technical specifications for developers, and a detailed explanation of the machine learning models used. Regular feedback sessions will be held with stakeholders to ensure alignment with project goals, and updates will be made as necessary based on their input.

5.1.2 Project Resources

1. **Backend Developers:** Responsible for developing the core processing components and APIs of the crop recommendation system. This includes handling the data preprocessing pipeline, implementing machine learning models for crop predictions, and managing the integration of IoT sensor data and weather APIs. The primary technologies used will be Python (for machine learning), Flask/Django for the back-end API, and libraries like Scikit-learn and TensorFlow for building the predictive models.
2. **Data Scientists/ML Engineers:** Focused on designing, training, and fine-tuning the machine learning models that power the crop recommendation system. They will work on developing predictive models based on environmental data, such as soil conditions, climate, and crop performance. Additionally, they will handle data analysis tasks and model evaluation to ensure accuracy and effectiveness in real-world agricultural settings.
3. **Frontend Developers:** Responsible for building the user interface where farmers can interact with the recommendation system. This will include the design of a dashboard to view personalized crop recommendations, weather information, and performance metrics. Modern web technologies like React.js and Bootstrap will be used to create an intuitive and user-friendly interface.
4. **Cloud Infrastructure and Database Management:** The system will rely on cloud services such as AWS or Google Cloud for hosting and scalability. A cloud database

solution (e.g., MySQL or PostgreSQL) will be used to store environmental data, user profiles, and recommendation outputs. These resources will ensure the system can handle large datasets and scale according to user demands.

5. Collaboration Tools: The development team will use GitHub for version control, tracking project progress, managing code contributions, and conducting code reviews. For team communication and collaboration, Google Meet and Slack will be used for real-time discussions, project updates, and brainstorming sessions.

5.2 Risk Management

Risk management is a critical component of our project to ensure its smooth development and deployment. By proactively identifying potential risks and devising mitigation strategies, we aim to minimize the negative impact on the project's timeline, functionality, and user satisfaction. This approach will contribute to the project's overall success and operational stability.

5.2.1 Risk Identification

During the risk identification process for our crop recommendation system, we have identified the following risks:

- **Technical Risk:**

1. **Data Integration Challenges:** Difficulties in integrating various data sources such as remote sensing data, IoT devices, and weather APIs, which may lead to inaccurate or incomplete recommendations.
2. **Model Performance Issues:** The potential for suboptimal performance in machine learning models, especially when dealing with diverse environmental data or when algorithms fail to capture complex interactions between soil conditions, climate, and crop profitability.
3. **Real-time Data Processing Delays:** Latency in processing realtime data from IoT sensors or weather stations, which may result in delayed or outdated crop recommendations.

- **Business Risk:**

1. **Budget Constraints:** Limited funding or resources may affect the implementation of advanced features such as the integration of IoT devices or the use of high-end machine learning models, potentially slowing down the development process.
2. **Shifting Market Demand:** Changes in agricultural trends or market preferences could lead to a misalignment between the system's recommendations and the needs of farmers, resulting in lower adoption or decreased effectiveness.
3. **Adoption Challenges:** Slow adoption by farmers or reluctance to trust the technology could hinder the system's widespread use, affecting the long-term sustainability of the project.

- **Security Risk:**

1. **Data Privacy Concerns:** The risk of unauthorized access to sensitive data, including user information or environmental data, leading to privacy violations.
2. **API Vulnerabilities:** Potential security weaknesses in the system's APIs used for data collection and communication, making it susceptible to exploitation or manipulation.
3. **Misuse of Recommendations:** The risk that incorrect or misleading crop recommendations, especially those not properly adjusted for specific regional conditions, could negatively impact farmers' financial outcomes and the credibility of the system.

5.2.2 Risk Analysis

After identifying potential risks, we conducted a thorough risk analysis to assess their impact and create effective mitigation plans:

- **Technical Risk:** To minimize risks related to data integration and machine learning model performance, we will ensure reliable access to external data sources, such as weather APIs. Regular model evaluations and updates will be implemented to maintain accuracy and adapt to evolving environmental factors. Additionally, scalable cloud infrastructure will be used to support real-time data processing and ensure the system can handle high usage during peak seasons.
- **Business Risk:** To address potential changes in market demands and user requirements, we will maintain clear communication with stakeholders throughout the project. The budget will be closely tracked, and project timelines will be adjusted to accommodate any unforeseen challenges. Engaging early with farmers and agricultural experts will ensure the platform meets user needs and increases adoption.
- **Security Risk:** Security of user data and recommendations will be prioritized by implementing encryption techniques and secure data storage practices. Regular security audits will be conducted to identify vulnerabilities, and strict access controls will be enforced for both user and environmental data. Moreover, measures will be taken to ensure the reliability of the recommendation system and prevent misuse of the platform for fraudulent purposes.

5.2.3 Overview of Risk Mitigation, Monitoring, and Management

1. Risk Mitigation:

- **Proactive Planning:** Developed and implemented contingency plans for each identified risk, ensuring that all stakeholders were aware of potential issues and their resolution strategies.
- **Resource Allocation:** Prioritized key resources (time, personnel, budget) to address critical risks first, ensuring adequate allocation for unforeseen challenges.
- **Third-party Collaboration:** Engaged early with third-party providers (for APIs or integrations) to ensure timely updates and collaboration, minimizing integration challenges.

2. Risk Monitoring:

- **Continuous Monitoring:** Used tools to continuously track the performance of the algorithms, user feedback, and system health, enabling early identification of any issues or performance degradation.
- **Real-time Alerts:** Set up real-time monitoring and alert systems for performance issues, security threats, or API malfunctions to detect and address them immediately.

- **Frequent Audits:** Scheduled regular security and performance audits to identify potential vulnerabilities and improve system performance.

3. Risk Management:

- **Dynamic Adaptation:** As the project progressed, the risk management plan was revisited regularly to adapt to new risks, incorporating feedback from the team and stakeholders.
- **Stakeholder Involvement:** Ensured regular communication with stakeholders to keep them updated on risk status, mitigation strategies, and any shifts in project scope or direction.
- **Post-Deployment Risk Review:** After the initial deployment, continued to monitor risk factors, especially in terms of market adoption and user satisfaction, refining the system based on real-world data and feedback.

These strategies helped minimize disruptions, maintained project timelines, and ensured the overall success of the project while safeguarding user and business interests.

5.3 Project Schedule

5.3.1 Project Task Set

- Conduct research and interviews with agricultural experts to understand the key factors influencing crop selection, such as soil composition, weather patterns, and market trends.
- Meet with project mentors to discuss the project goals, scope, and expectations for the crop recommendation system.
- Define the problem statement and gather detailed requirements from relevant research papers, online resources, and expert consultations.
- Brainstorm potential solutions, focusing on integrating environmental data and machine learning models for accurate crop recommendations.
- Document the problem statement, research findings, and initial system design for future reference.
- Identify the technology stack and tools required for the implementation, such as machine learning libraries, weather data APIs, and web development tools.
- Research machine learning algorithms suitable for crop recommendation, including decision trees, random forests, and hybrid models.
- Assign roles to team members based on their expertise in areas such as data collection, model development, and user interface design.
- Familiarize the team with the necessary tools and technologies, including Python, scikit-learn, and web development frameworks.
- Design the basic user interface for the platform, enabling users to input environmental data and receive crop recommendations.
- Finalize the system architecture and design, ensuring smooth integration between the front-end, back-end, and machine learning components.
- Begin developing the core components of the system, including data collection, preprocessing pipelines, and integration with external weather data sources.
- Select and test machine learning models for crop recommendation, evaluating their predictive accuracy using sample datasets. Develop and integrate the chatbot module for addressing frequently asked questions and assisting users.
- Perform internal testing to ensure that all modules work together without issues, focusing on system functionality and usability.
- Deploy the system on a local or cloud-based server for testing and demonstration purposes.
- Gather feedback from mentors and potential users to assess the platform's performance, accuracy, and ease of use.

- Refine the system based on feedback, adjusting models and interfaces to improve performance.
- Conduct final system testing to ensure all components meet the project's requirements and function as expected.
- Document the entire development process, including the design, codebase, testing results, and challenges encountered.
- Prepare the final project report, presentation, and code for submission, and present the system for final evaluation.

5.3.2 Timeline Chart

Figure 5.1 is the summarized project timeline chart. It briefly explains the schedule of our entire project from inception to projected completion.

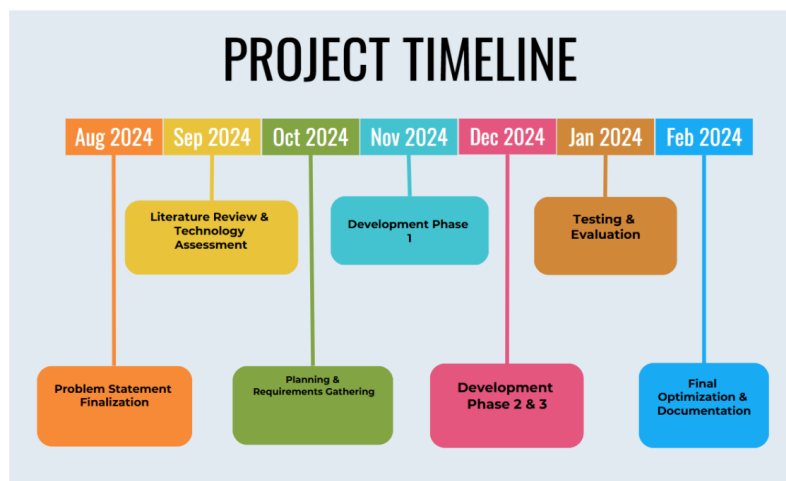


Figure 5.1: Summarized Timeline Chart

5.4 Team Organization

5.4.1 Team structure

- **Rahul Kewalramani:** Backend, Database Operations ML Developer.
- **Saurabh Rajpoot:** Frontend ML developer.
- **Prof. Hema Kumbhar:** Internal mentor and guide.

5.4.2 Management reporting and communication

- **Management:** To manage the crop recommendation project efficiently, we adopted a structured approach using collaborative tools. Google Drive was used to organize and share documents, research papers, and datasets among team members for easy access. For code management and version control, we used Git and GitHub, facilitating smooth collaboration and tracking development changes. Weekly meetings were held to review progress, discuss obstacles, and plan upcoming tasks, ensuring alignment with project goals.
- **Reporting:** We followed a weekly reporting protocol to update our internal mentor, Prof. Hema Kumbhar, on project milestones. These reports included progress on model development, data integration challenges, and proposed solutions. The mentor's feedback guided our adjustments and helped maintain focus on project objectives. We also shared periodic progress updates with the department's review panel to meet academic standards.

- **Communication:** For team communication, we primarily relied on Slack and Google Meet for updates, discussions, and brainstorming sessions. Scheduled weekly meetings with the mentor ensured consistent feedback and support. This communication strategy enabled the team to address challenges promptly and stay updated on project advancements.

This organized approach to management, reporting, and communication ensured transparency, accountability, and efficiency, ultimately contributing to the successful progression of the project.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

- **User Authentication Module:** Function: Allows users (farmers, researchers) to securely register and log in to the system.
- **Soil and Environmental Data Input Module:** Function: Collects user-input or sensor-based environmental data including:
 - **Soil composition (NPK values):** Temperature, humidity, rainfall, pH level
- **Weather Forecast Report Module:** Function: Provides real-time and forecasted weather information.
- **Crop Recommendation Module:** Function: Recommends optimal crops based on: 1. Environmental conditions 2. Soil health 3. Historical yield data
- **Yield Prediction Module:** Function: Predicts expected yield for selected crops using historical and environmental data.
- **Visualization Dashboard Module:** Function: Displays outputs, trends, and data insights in a user-friendly manner.
- **Admin Panel (Optional):** Function: Allows administrators to manage users, update datasets, and monitor system logs.

6.2 Tools and Technologies Used

- **Languages:** Python for backend and model development, JavaScript (React) for frontend.
- **Libraries:** Pandas, NumPy, Matplotlib, Scikit-learn.
- **Models:** Random Forest Regressor, Support Vector Machine, K-Nearest Neighbors (KNN).
- **Frameworks:** Flask and Bootstrap .
- **Frontend Tools:** HTML + CSS + JavaScript.
- **Database:** SQLite – Version control and project collaboration. for storing user data, queries, and interactions.
- **Containerization:** Docker for modular deployment.
- **Deployment:** GitHub – Version control and project collaboration.

6.3 Algorithm Details

6.3.1 Random Forest Regressor

- Ensemble method using multiple decision trees to predict continuous output.
- Why used? Robust to noise, handles nonlinear relationships well.
- Performance: Achieved 91
- Evaluation Metrics: MAE (Mean Absolute Error), RMSE (Root Mean Squared Error).

6.3.2 Support Vector Machine (SVM)

- Constructs a hyperplane in high-dimensional space to separate crop classes.
- Why used? Effective for complex and high-dimensional datasets.
- Kernel Used: RBF (Radial Basis Function).

6.3.3 K-Nearest Neighbors (KNN)

- Classifies data points based on the majority class of their nearest neighbors.
- Why used? Works well for small datasets with simple decision boundaries.
- Tuning: K-value selected using cross-validation.
- Metric Used: Accuracy, Precision.

CHAPTER 7

SOFTWARE TESTING

7.1 Software Testing

Software testing is the process of evaluating a system or its components with the intent to determine whether it meets the specified requirements. The goal is to find any defects or bugs in the software that could hinder its functionality, performance, or user experience. In our project, the testing process is critical to ensure that all functionalities, including user interactions, data inputs, soil analysis, and clustering, perform as expected.

The major types of testing performed in this project include:

7.1.1 Unit Testing

Unit testing involves testing individual units or components of the software in isolation to ensure they work correctly. Each function or method is tested in its most basic form. Developers typically conduct unit tests during the development phase, using tools like JUnit or PyTest.

7.1.2 Integration Testing

Integration testing is performed after individual units are tested. It focuses on evaluating the interaction and data flow between different modules or components. The purpose is to detect defects that may arise when the modules work together.

7.1.3 System Testing

System testing is a high-level test conducted on the complete integrated system. It ensures that the system complies with all the specified requirements. This includes both functional testing (like checking the video comment system) and non-functional testing (like security and performance).

7.1.4 Functional Testing

Functional testing verifies that the software behaves according to the defined functional requirements. It ensures that the system implements the required functionality correctly, such as input validation, business rules, and output generation.

7.1.5 Acceptance Testing (User Acceptance Testing - UAT)

Acceptance testing is the final level of testing, conducted to ensure that the software meets the business needs and end-user requirements. It is typically performed by stakeholders or clients before the software is released.

7.2 Test cases & Test Results

Test Case ID	Test Scenario	Expected Output	Actual Output
TC_USER_001	User Login with valid credentials	User is logged in and redirected to the homepage	User is logged in and redirected to the homepage
TC_USER_002	User signup with valid details	User account is created, and user is redirected to the homepage	User account is created, and user is redirected to the homepage
TC_USER_003	User login with invalid credentials	Error message: “Invalid username or password” is displayed	Error message: “Invalid username or password” is displayed
TC_ADMIN_001	View a user’s own videos in the dashboard	All uploaded data are displayed along with user data	All uploaded inputs are displayed along with user data
TC_SEARCH_001	Search for a query	The relevant comment(s) are displayed in the search results	The relevant comment(s) are displayed in the search results
TC_SEARCH_002	Search with no matching results	Message: “No data found” is displayed	Message: “No data found” is displayed
TC_SEARCH_003	Search with partial query match	Inputs with a matching partial title are displayed	Inputs with a matching partial title are displayed

Table 7.1: Test Cases

CHAPTER 8

RESULTS

8.1 Project Outcomes

Our project successfully achieved its goals by leveraging machine learning and real-time data to assist farmers in making informed, data-driven agricultural decisions. Below are the key outcomes:

- **Accurate Crop Recommendation System:** Developed and deployed a machine learning-based crop recommendation engine.
Achieved 95.2
Tailored suggestions based on soil health, weather, and market trends.
- **Yield Prediction Module:** Implemented regression models (Random Forest and XGBoost) to estimate crop yield.
Reached prediction accuracy up to 91.6
Enabled farmers to forecast production and plan accordingly.
- **Weather Integration:** Integrated weather API to fetch live and forecasted data (temperature, humidity, rainfall).
Helped users plan irrigation, sowing, and harvesting based on climate trends.
- **Profitability Insights:** Provided basic market-driven profitability analysis by factoring in crop prices.
Supported economically optimized crop selection, not just agriculturally viable ones.
- **User-friendly Web Application:** Designed and built a responsive, intuitive frontend using HTML, CSS, and Bootstrap.
Enabled secure user login, dashboard access, and real-time interaction.
Displayed predictions and recommendations through clean visual interfaces and charts.

8.2 Screen Shots

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/register/". The page title is "CropRecommendationSystem". The navigation bar includes links for "Home", "About", and "Contact", along with buttons for "Signup", "Signin", and "Admin Signin". The main content area is titled "User Registration" and contains a form with the following fields: "First name" (with a placeholder "First Name"), "Last name" (with a placeholder "Last Name"), "Email" (with a placeholder "Email"), "Password" (with a placeholder "Password"), "Image" (with a "Browse..." button and "No file selected." text), "Mobile" (with a placeholder "Mobile"), and "Address" (with a placeholder "Address in detail"). A green "Register" button is located at the bottom of the form. The footer section includes the text "CropRecommendation", "The Company", and "International sites".

Figure 8.1: Result 1

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/signin/". The page title is "CropRecommendationSystem". The navigation bar includes links for "Home", "About", and "Contact", along with buttons for "Signup", "Signin", and "Admin Signin". The main content area is titled "User Sign In" and contains a form with the following fields: "Email" (with a placeholder "Email") and "Password" (with a placeholder "Password"). A green "Sign in" button is located at the bottom of the form. The footer section includes the text "CropRecommendation", "The Company", and "International sites". Under "The Company", there are links for "Site Map", "Legal", "Agent Admin", "Careers", "Affiliate", and "Privacy Policy". Under "International sites", there are links for "Amazon", "Flipkart", "Croma", "Gadgets Now", and "Reliance Digital".

Figure 8.2: Result 2

The screenshot shows the home page of the CropRecommendationSystem. The navigation bar includes links for Home, Prediction, History, Change Password, and Logout, along with a 'Welcome, sachin' message. The main section is the 'Crops Prediction Form', which contains input fields for Nitrogen in soil (Ratio of nitrogen in soil), Phosphorous in soil (Ratio of phosphorous in soil), Potassium in soil (Ratio of potassium in soil), Temperature (in °C), Relative Humidity, pH value of soil, Humidity, and Rainfall (in mm). A green 'Predict' button is located at the bottom of the form. The footer section includes 'CropRecommendation' with a placeholder text, 'The Company' with links to Site Map and Legal, and 'International sites' with links to Amazon and Flipkart.

Figure 8.3: Result 3

The screenshot shows the 'My History' page of the CropRecommendationSystem. It features a table with columns for #, Search Date, Output, Search Date, and Action. The table contains one entry with the following data:

#	Search Date	Output	Search Date	Action
7	[Nitrogen: [70], phosphorous: [7], potassium: [47], temperature: [28.45], humidity: [94.72], pH: [6.18], rainfall: [26.30]]	Muskimelon	June 15, 2023, 4:27 p.m.	Delete

Below the table, it says 'Showing 1 to 1 of 1 entries'. The footer section is identical to the one in Figure 8.3, including links for 'The Company' (Site Map, Legal, Agent Admin, Careers, Affiliate, Privacy Policy) and 'International sites' (Amazon, Flipkart, Croma, Gadgets Now, Reliance Digital).

Figure 8.4: Result 4

CHAPTER 9

CONCLUSIONS

9.1 Conclusion

his project demonstrated the significant impact of data-driven technologies, particularly machine learning, in enhancing modern agricultural practices. The web-based crop recommendation system that was developed extended beyond the capabilities of traditional systems by incorporating both environmental parameters—such as soil composition, temperature, humidity, and rainfall—and economic factors, including market demand and crop profitability.

By suggesting alternative crops based on real-time data and market trends, the system enabled farmers to make more informed decisions aimed at optimizing both crop productivity and financial returns. Machine learning models played a pivotal role in this process, efficiently analyzing large and complex datasets to deliver precise, condition-specific recommendations.

The system's adaptability allowed it to evolve and improve over time, dynamically responding to shifting environmental and market conditions. As a result, it provided not only accurate insights but also a scalable and sustainable approach to farming. In the long term, the application of such web-based intelligent systems holds the potential to promote sustainable agriculture by maximizing resource efficiency, reducing waste, increasing profitability, and empowering farmers with actionable, data-driven insights.

9.2 Future Work

1. **Integration of Diverse Data Sources:** Future improvements will focus on adding remote sensing data, like satellite imagery, to capture soil moisture, crop health, and land use. Combined with real-time weather updates and IoT devices (e.g., soil sensors), this integration will enable precise, location-based recommendations that adjust to changing conditions.
2. **Enhancing Predictive Accuracy:** To improve recommendation accuracy, future versions will explore advanced models, including hybrid approaches combining decision trees, neural networks, and gradient boosting. Deep learning techniques will also be tested to better capture complex relationships between environmental factors and economic outcomes.
3. **IoT Integration:** Incorporating IoT devices like soil and weather sensors will allow for real-time data collection on nutrients, pH, and moisture, providing highly localized and timely recommendations to support informed decision-making.

9.3 Applications

1. **Social Media Engagement:** The platform assists content creators by streamlining responses to frequently asked questions and clustering similar comments, improving engagement and allowing for more efficient interaction with their audience.
2. **Customer Support Automation:** Query matching automate the handling of repetitive user inquiries, making it ideal for businesses looking to enhance customer support through quicker and more accurate responses to common questions.

3. **Multilingual Content Moderation:** Future enhancements could allow the system to be applied across platforms with multilingual audiences, making it a powerful tool for moderating and responding to comments in diverse linguistic environments.
4. **Content Categorization:** The comment clustering feature can be used in e-commerce and review platforms to categorize customer feedback, allowing companies to address specific concerns such as product quality, pricing, or delivery issues more effectively.
5. **Precision Agriculture Support:** Lays the foundation for integration with IoT sensors and drones to enable more accurate, real-time farm management.
Can support future expansion into pest prediction, fertilizer suggestion, or disease detection.
6. **Agricultural Extension Services:** Useful for government agencies and NGOs providing advisory services to farmers.
Can be integrated into rural development programs to assist smallholder and marginal farmers.
7. **Educational Research Use:** Acts as a case study for the integration of AI/ML in agriculture.
Useful for academic purposes in agricultural technology, data science, and environmental science fields.
8. **Mobile or Web Application Platforms:** Can be scaled into a mobile app for wider adoption by farmers.
Integration with multilingual support can help reach rural and non-English-speaking users.

CHAPTER 10

APPENDIX

10.1 Appendix A

Technical Feasibility

- The project leverages advanced technologies, including machine learning algorithms and predictive modeling, ensuring both technical feasibility and scalability for the crop recommendation platform.
- Utilizing Python-based machine learning libraries and a cloud-hosted environment, the system can efficiently process and analyze environmental data, providing reliable crop suggestions. This technical framework supports high availability and scalability, accommodating the growing demand for agricultural insights and ensuring smooth, continuous operation for end-users.

Operational Feasibility

- User testing and feedback from farmers and agricultural advisors indicate that the crop recommendation platform is operationally feasible.
- The integration of environmental data analysis with machine learning models has proven effective in suggesting suitable crops, enhancing decision-making processes. The platform's ability to provide tailored crop recommendations based on real-time conditions has been positively received, confirming its operational viability.

Economic Feasibility

- A cost-benefit analysis was conducted to evaluate the economic feasibility of the proposed crop recommendation platform.
- The expenses involved in developing, deploying, and maintaining the platform were compared with the expected benefits, including increased crop yield, optimized resource usage, and improved profitability for farmers.
- The analysis showed that the platform is economically sustainable, with the potential benefits significantly exceeding the costs, making it a practical solution to enhance agricultural productivity and farmer profitability.

10.2 Appendix B

10.2.1 Survey Paper Details

Paper Status: Published

Name of Conference: 2025 1st International Conference on AIML-Applications for Engineering Technology (ICAET).

Page Numbers: 7

DOI:10.1109/ICAET63349.2025.10932168



Figure 10.1: Conference Certificate



Figure 10.2: Conference Certificate

10.2.2 Event Participation Details

Event Name: Impetus & Concepts (INC)

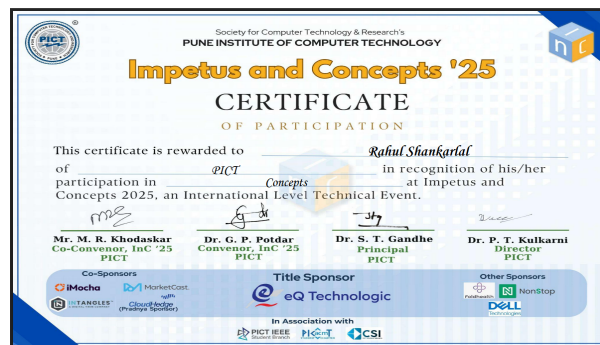


Figure 10.3: Rahul's Inc Certificate



Figure 10.4: Saurabh's Inc Certificate

10.3 Appendix C

10.3.1 Plagiarism Report

Overall 16 received.

Applied for the conference; yet to receive acceptance

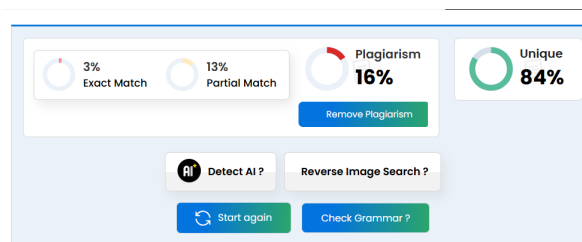


Figure 10.5: Plagiarism Report Result

CHAPTER 11

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